HW 3

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$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \log\left(\frac{P(Y_i=1 \mid X_i)}{1-P(Y_i=1 \mid X_i)}\right) = X_i^T\beta$$

Solve for π_i:

$$\frac{\pi_i}{1-\pi_i} = \exp(X_i^T\beta)$$

$$\pi_i = \exp(X_i^T\beta) - \pi_i \exp(X_i^T\beta)$$

$$\pi_i + \pi_i \exp(X_i^T\beta) = \exp(X_i^T\beta)$$

$$\pi_i(1+\exp(X_i^T\beta)) = \exp(X_i^T\beta)$$

$$\pi_i = \frac{\exp(X_i^T\beta)}{1+\exp(X_i^T\beta)}$$

$$P(\vec{Y}=y \mid X) = \prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i} = L(\pi_i(\beta) \mid \vec{y})$$

$$\ell(\beta;\vec{y}) = \sum_{i=1}^{n}\left[\log(\pi_i^{y_i}) + \log(1-\pi_i)^{(1-y_i)}\right]$$

$$= \sum_{i=1}^{n} y_i \log(\pi_i) + (1-y_i)\log(1-\pi_i)$$

$$= \sum_{i=1}^{n} y_i \log\left(\frac{\exp(X_i^T\beta)}{1+\exp(X_i^T\beta)}\right) + (1-y_i)\log\left(\frac{1}{1+\exp(X_i^T\beta)}\right)$$

$$\frac{\partial}{\partial\beta_j}\left(y_i \log\left(\frac{\exp(X_i^T\beta)}{1+\exp(X_i^T\beta)}\right)\right)$$

$$= y_i \frac{\partial}{\partial\beta_j}\left(X_i^T\beta - \log(1+\exp(X_i^T\beta))\right)$$

$$= y_i\left(X_{ij} - \frac{\partial}{\partial\beta_j}\log(1+\exp(X_i^T\beta))\right)$$

$$= y_i\left(X_{ij} - \frac{1}{1+\exp(X_i^T\beta)}\exp(X_i^T\beta)\cdot X_{ij}\right)$$

$$= y_i X_{ij}(1-\pi_i)$$

$$(1-y_i)\frac{\partial}{\partial\beta_j}\log\left(\frac{1}{1+\exp(X_i^T\beta)}\right)$$

$$= (1-y_i)\frac{\partial}{\partial\beta_j}\left(\log 1 - \log(1+\exp(X_i^T\beta))\right)$$

$$= -(1-y_i)\frac{\partial}{\partial\beta_j}\left(\log(1+\exp(X_i^T\beta))\right)$$

$$= -(1-y_i)\frac{1}{1+\exp(X_i^T\beta)}\cdot\exp(X_i^T\beta)\cdot X_{ij}$$

$$= -(1-y_i)X_{ij}\pi_i$$

$$\frac{\partial\ell}{\partial\beta_j} = \sum_{i=1}^{n} y_i X_{ij} - \cancel{\pi_i y_i X_{ij}} - X_{ij}\pi_i + \cancel{\pi_i y_i X_{ij}}$$

$$= \sum_{i=1}^{n} X_{ij}(y_i - \pi_i)$$

<u>Hessian:</u>

$$\frac{\partial^2\ell}{\partial\beta_j\partial\beta_k} = \frac{\partial}{\partial\beta_k}\left(\sum_{i=1}^{n} X_{ij}(y_i - 1 - (1-\pi_i))\right)$$

$$= \sum_{i=1}^{n} x_{ij}\left(\cancel{\frac{\partial}{\partial\beta_k} y_i} - \cancel{\frac{\partial}{\partial\beta_k} 1} - \frac{\partial}{\partial\beta_k}(1-\pi_i)\right) \qquad \frac{\partial}{\partial\beta_k}\frac{1}{1+\exp(x_i^T\beta)} = \frac{x_{ik}\exp(x_i^T\beta)}{(1+\exp(x_i^T\beta))} = x_{ik}\pi_i$$

$$= -\sum_{i=1}^{n} x_{ij}x_{ik}\pi_i$$

<u>Newton Update</u>

$$\beta_{t+1} = \beta_t + \Big[\sum_{i=1}^{n} x_{ij}x_{ik}\pi_i\Big]^{-1}_{j,k\in\{1,\dots,p\}} \Big[\sum_{i=1}^{n} x_{ij}(y_i-\pi_i)\Big]_{j\in\{1,\dots,p\}}$$

<u>Convexity</u>

We show the Hessian $\nabla^2 \ell = \Big[-\sum_{i=1}^{n} x_{ij}x_{ik}\pi_i\Big]_{i,j}$ is positive semi-definite. Each summand $x_{ij}x_{ik}\pi_i$ can be written as $x_j^{*T}x_k^{*}$ where

$x_j^{*T} = [\sqrt{\pi_1}\,x_{1j}, \sqrt{\pi_2}\,x_{2j}, \dots, \sqrt{\pi_n}\,x_{nj}]$, $x_k^{*T} = [\sqrt{\pi_1}\,x_{1k}, \dots, \sqrt{\pi_n}\,x_{nk}]$

because $\pi_i \geq 0 \ \forall \beta$.

The Hessian $\nabla^2 \ell$ may be written as

$$-(\underset{p\times n}{X^{*T}}\underset{n\times p}{X^{*}}) = -\begin{bmatrix} \sqrt{\pi_1}\,x_{11} & \sqrt{\pi_2}\,x_{21} & \cdots & \sqrt{\pi_n}\,x_{n1} \\ \sqrt{\pi_1}\,x_{12} & & & \\ \vdots & & & \\ \sqrt{\pi_1}\,x_{1,(p+1)} & & & \sqrt{\pi_n}\,x_{n,(p+1)} \end{bmatrix} \begin{bmatrix} \sqrt{\pi_1}\,x_{11} & \sqrt{\pi_1}\,x_{n1} & \cdots & \sqrt{\pi_1}\,x_{1,(p+1)} \\ \sqrt{\pi_2}\,x_{21} & \sqrt{\pi_2}\,x_{22} & & \\ \vdots & & & \\ \sqrt{\pi_n}\,x_{n1} & & \cdots & \sqrt{\pi_n}\,x_{n,(p+1)} \end{bmatrix}$$

$$\overset{\checkmark}{=} -\begin{bmatrix} \sum_{i=1}^{n}\pi_i x_{i1}^2 & \sum_{i=1}^{n}\pi_i x_{i1}x_{i2} & \cdots & \sum_{i=1}^{n}\pi_i x_{i1}x_{i(p+1)} \\ \vdots & \ddots & & \vdots \\ \sum \pi_i x_{i,(p+1)}x_{i,1} & \cdots & & \sum_{i=1}^{n}\pi_i x_{n,(p+1)}^2 \end{bmatrix}$$

Which is negative-semi definite because $X^{*T}X^{*}$ is a quadratic form therefore PSD. Thus $-\nabla^2\ell$ is PSD and the score function $\ell(\theta)$ is anti-convex, so $-\ell(\theta)$ is convex and may be minimized accordingly.

<u>Problem 2: MM:</u>

a) <u>WTS:</u> $-\log(1+\exp(X_i^T\theta)) \geq -\log(1+\exp(X_i^T\theta^{(k)})) - \dfrac{\exp(X_i^T\theta)-\exp(X_i^T\theta^{(k)})}{(1+\exp(X_i^T\theta^{(k)}))}$

Let $u = 1+\exp(X_i^T\theta)$, $v = 1+\exp(X_i^T\theta^{(k)})$

$-\log(\cdot)$ is a convex function, so by first order approx.

$$-\log(u) \geq -\log(v) - \nabla\log(v)^T(u-v)$$

$$\nabla\log(v) = \frac{1}{v}$$

$$-\log(u) \geq -\log(v) - \frac{u-v}{v}$$

$$\Leftrightarrow \quad -\log(1+\exp(X_i^T\theta)) \geq -\log(1+\exp(X_i^T\theta^{(k)})) - \frac{\exp(X_i^T\theta) - \exp(X_i^T\theta^{(k)})}{1+\exp(X_i^T\theta^{(k)})}$$

as desired: when $\theta = \theta^{(k)}$, $\exp(X_i^T\theta) - \exp(X_i^T\theta^{(k)}) = 0$,

so $-\log(1+\exp(X_i^T\theta)) \geq -\log(1+\exp(X_i^T\theta^{(k)})) = -\log(1+\exp(X_i^T\theta))$

gives equality.

b) Want to find $g(\cdot)$ s.t.

$g(\theta \mid \theta^{(t)}) \leq \ell(\theta) \quad \forall \theta \in \Theta$ and

$g(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)})$

$$\ell(\theta) = \sum_{i=1}^n y_i \log\left(\frac{\exp(X_i^T\beta)}{1+\exp(X_i^T\beta)}\right) + (1-y_i)\log\left(\frac{1}{1+\exp(X_i^T\beta)}\right)$$

$$= \sum y_i\left[\log(\exp X_i^T\beta) - \log(1+\exp(X_i^T\beta))\right] - (1-y_i)\log(1+\exp(X_i^T\beta))$$

$$= \sum y_i X_i^T\beta - y_i\log(1+\exp(X_i^T\beta)) + y_i\log(1+\exp(X_i^T\beta)) - \log(1+\exp(X_i^T\beta))$$

$$= \sum_{i=1}^n y_i X_i^T\beta - \log(1+\exp(X_i^T\beta))$$

$$\geq \sum_{i=1}^n y_i X_i^T\beta - \log(1+\exp(X_i^T\beta^{(k)})) - \frac{\exp(X_i^T\beta) - \exp(X_i^T\beta^{(k)})}{1+\exp(X_i^T\beta^{(k)})}$$

$$= \sum_{i=1}^n y_i X_i^T\beta + \sum_{i=1}^n\left[\frac{\exp(X_i^T\beta^{(k)})}{1+\exp(X_i^T\beta^{(k)})} - \log(1+\exp(X_i^T\beta^{(k)})) - \frac{\exp(X_i^T\beta)}{1+\exp(X_i^T\beta^{(k)})}\right]$$

$$\exp(X_i^T\beta^{(k)}) = (\exp(pX_i^T\beta^{(k)}))^{1/p}$$

$$= \left(\prod_{j=1}^p \exp(pX_{ij}\beta^{(k)})\right)^{1/p} \leq \frac{1}{p}\sum_{j=1}^p \exp(pX_{ij}\beta_j^{(k)})$$

$$\geq \sum_{i=1}^n y_i X_i^T\beta + \sum_{i=1}^n \frac{\exp(X_i^T\beta^{(k)})}{1+\exp(X_i^T\beta^{(k)})} - \log\Big(1+\underbrace{\frac{1}{p}\sum_{j=1}^p \exp(pX_{ij}\beta_j^{(k)})}_{:=S}\Big) - \frac{\frac{1}{p}\sum_{j=1}^p \exp(pX_{ij}\beta_j)}{1+\frac{1}{p}\sum \exp(pX_{ij}\beta_j^{(k)})}$$

$$-\log(1+S^{(k)}) - \frac{S}{1+S^{(k)}}, \quad S > 0$$

$\geq -S^{(k)}$

c) $$\frac{\partial}{\partial \beta_\ell}\left[-\frac{1}{\rho}\sum_{i=1}^{n}\frac{\exp(X_i^T\beta^{(k)})}{1+\exp(X_i^T\beta^{(k)})}\sum_{j=1}^{p}\exp(\rho X_{ij}(\beta_j-\beta_j^{(k)}))+\sum_{i=1}^{n}Y_i X_i^T\beta\right]$$

$$=\frac{\partial}{\partial \beta_\ell}\left[-\frac{1}{\rho}\sum_{i,j}\frac{\exp(X_i^T\beta^{(k)})}{1+\exp(X_i^T\beta^{(k)})}\cdot\exp(\rho X_{ij}\beta_j)\cdot\exp(-\rho X_{ij}\beta^{(k)})\right]+\sum_{i=1}^{n}Y_i X_{i\ell}$$

All other j terms $\to 0$

$$=-\frac{1}{\cancel{\rho}}\sum_{i=1}^{n}\frac{\exp(X_i^T\beta^{(k)})}{1+\exp(X_i^T\beta^{(k)})}\cdot\cancel{\rho}X_{i\ell}\cdot\exp(\rho X_{i\ell}\beta_\ell)\cdot\exp(-\rho X_{i\ell}\beta^{(k)})+\sum_{i=1}^{n}Y_i X_{i\ell}$$

which equals the given expression. Setting it equal to 0 finds a global maximum because $-g(\beta|\beta_0)$ is convex (sum of a linear function and $a\exp(z)$) in β.

Problem 4:

E-Step

$t_1, \ldots, t_n \sim Exp(\lambda)$

We observe

$y_1, \ldots, y_n$ where $y_i = \min(t_i, C_i) \leftarrow$ subj. i censored at time C_i

$\delta_i = 1$ if $y_i \le c_i$ (if not censored)

$\delta_i = 0$ else

<u>We don't know:</u>

Survival times for censored individuals:

$t_i = \delta_i y_i + (1-\delta_i) z_i$

where z_i is the true unobserved survival time.

$z_i = c_i + w_i$, where w_i is censored additional survival time.

<u>Likelihood</u>: $L(\lambda; \vec{t}) = \prod_{i=1}^{n} f(x_i; \lambda)$

$= \prod_{i=1}^{n} \frac{1}{\lambda} \exp\left(\frac{-t_i}{\lambda}\right) = \frac{1}{\lambda^n} \exp\left(-\frac{1}{\lambda} \sum_{i=1}^{n} t_i\right)$

$= \frac{1}{\lambda^n} \exp\left(-\frac{1}{\lambda} \sum_{i=1}^{n} \delta_i y_i + (1-\delta_i) z_i\right)$

$\ell(\lambda; t) = -n\log(\lambda) - \frac{1}{\lambda} \sum_{i=1}^{n} [\delta_i y_i + (1-\delta_i) z_i]$

is the log-likelihood of the full data density.

<u>Q-function:</u>

$E_z[\ell(\lambda; t) \mid t, \lambda]$

$= E_z\left[-n\log(\lambda) - \frac{1}{\lambda} \sum_{i=1}^{n} [\delta_i y_i + (1-\delta_i) z_i]\right]$

$= -n\log(\lambda) - \frac{1}{\lambda} \sum_{i=1}^{n} \delta_i y_i - \frac{1}{\lambda} \sum_{i=1}^{n} (1-\delta_i) E(z_i)$

Find $E(z_i)$:

$E(z_i) = E(c_i + w_i) = c_i + E(z_i \mid z_i \ge c_i)$

By memoryless property of exponentials, the additional survival time $(z_i \mid z_i \ge c_i) \sim Exp(\lambda)$ and has $E(w_i) = \lambda$.

So $E(z_i) = c_i + \lambda$.

Then $Q(\lambda \mid \lambda_0) = -n\log(\lambda) - \frac{1}{\lambda}\left[\sum_{i=1}^{n} \delta_i y_i + (1-\delta_i)(c_i + \lambda_0)\right]$

<u>M-Step:</u>

$\frac{\partial}{\partial \lambda}\left[-n\log(\lambda) - \frac{1}{\lambda}\left[\sum_{i=1}^{n} \delta_i y_i + (1-\delta_i)(c_i + \lambda_0)\right]\right]$

$= \frac{-n}{\lambda} + \frac{1}{\lambda^2}\left[\sum \delta_i y_i + (1-\delta_i)(c_i + \lambda_0)\right] := 0$

$\Rightarrow -n\lambda + \sum \delta_i y_i + (1-\delta_i)(c_i + \lambda_0)] = 0$

$\Rightarrow \hat{\lambda} = \frac{\sum_{i=1}^{n}[\delta_i y_i + (1-\delta_i)(c_i + \lambda_0)]}{n}$